

DATA
GOVERNANCE
IN AFRICA



DATA GOVERNANCE IN AFRICA PROGRAMME

AI4STARTUPS DATA & REGULATORY LANDSCAPE REPORT

MARCH 2026

INTRODUCTION

About the AI4Startups Programme and Data Governance in Africa Initiative

AI4Startups is a programme focused on strengthening the African tech ecosystem by fostering AI development, responsible data governance, data sharing and innovation with a particular emphasis on the agricultural sector. The initiative is implemented by Digital Africa, in collaboration with Expertise France, as part of the broader Data Governance in Africa Initiative.

Data Governance in Africa is a joint initiative supported by the African Union (AU) and the European Union (EU), implemented by a consortium of European partners including Germany (GIZ), Belgium (Enabel), France (Expertise France), Estonia (ESTDEV), Finland (HAUS), and Digital Africa. It forms part of the Team Europe and Global Gateway strategy, aiming to help African nations harmonise data policies and regulations while unlocking the potential of AI-driven solutions. The programme is funded with €60 million from the European Union and five of its Member States, running from January 2023 to July 2026.

In this context, the AI4Startups programme supports African startups through hackathons, mentorship programmes, and technical and legal capacity building particularly for data-dependent, innovation-led ventures in agriculture, food systems, climate, and energy. Its activities span the eight priority countries at the heart of this study: Ghana, Senegal, Côte d'Ivoire, Kenya, Rwanda, Tanzania, South Africa, and Botswana.

About This Study

This Data & Regulatory Landscape Study was conducted by Cage Patrick & Company Advocates as part of the AI4Startups programme's requirements on regulatory insights from startup experiences. The study was commissioned to help Digital Africa better understand the legal needs and regulatory challenges faced by African startups operating in data-intensive sectors, and to generate actionable insights that will inform the development of practical legal frameworks and support tools.

The study does not represent the views of Digital Africa or Expertise France, and the views expressed also do not include those of Cage Patrick & Company Advocates, as its involvement was limited to the design, conduct, and analysis of this data landscape survey. The findings and recommendations in this report are those of the study team, grounded in primary data collected from 39 survey respondents and 12 qualitative interviews across the eight priority countries.

The legal and regulatory dimensions of data use remain a significant barrier for many African startups particularly in sectors like agritech and food systems, where qualitative data are not easily available and sensitive commercial and environmental data are frequently involved. These barriers are compounded by a lack of adapted legal instruments, regulatory ambiguity, and limited access to legal expertise. This study sets out to document those barriers through the experiences of the startups themselves, and to translate those experiences into concrete, actionable recommendations.

METHODOLOGY

Research Design

This study employed a two-phase mixed-methods design, combining a quantitative survey with qualitative in-depth interviews. The two phases were designed to be complementary: the survey captured the scale and distribution of regulatory and data governance challenges across a broad cohort, while the interviews provided the narrative depth and contextual specificity needed to understand why and how those challenges manifest in practice.

Parameter	Detail
Survey respondents	39 (in-scope, deduplicated)
Qualitative interviews	12
Geographic scope	Ghana, Senegal, Côte d'Ivoire, Kenya, Rwanda, South Africa, Tanzania, Botswana
Sector scope	AgriTech, Climate Tech, Energy Tech, Food Tech
Survey platform	Youform & Typeform (CSV export)
Interview format	Virtual (Zoom, Google Meet, phone) – 30–45 minutes
Languages	English

Phase 1: Quantitative Survey

A structured online survey of 20 questions was distributed to African startups operating in the eight priority countries and across four priority sectors. The survey was distributed via the Youform and Typeform platforms and used conditional branching logic to ensure respondents only answered questions relevant to their profile.

The survey was structured across five thematic sections. Section 1 (Questions 1–6) covered consent and organisational profile, capturing the startup name, countries of operation, main sector, and funding stage. Section 2 (Questions 7–11) addressed data and AI usage, including a five-point data criticality scale (Q7), current AI/ML status (Q8), an optional conditional question on AI adoption barriers for active AI users (Q9), data access challenges (Q10), and whether the startup had previously shared data with another organisation (Q11). Section 2A (Question 12) was a conditional follow-up on data sharing challenges, shown only to respondents who answered “Yes” to Q11. Section 3 (Questions 13–17) covered the regulatory landscape, including a country-level compliance difficulty matrix across all eight priority countries (Q13), whether data protection laws primarily help or limit operations (Q14), the most difficult regulatory areas (Q15, up to three selections), legal support capacity (Q16), and the approach to data-sharing agreements (Q17). Section 4 (Questions 18–20) addressed support needs and follow-up, asking respondents to identify their most needed resources (Q18), whether they were interested in a follow-up interview (Q19), and an optional open-ended field for additional comments (Q20).

A total of 39 in-scope responses were collected and included in this analysis. Responses from startups outside the designated geographic or sectoral scope were excluded from the primary analysis. One duplicate submission was identified and removed.

Phase 2: Qualitative Interviews

Twelve semi-structured qualitative interviews were conducted with startups that had indicated interest in follow-up participation and met the selection criteria: significant data dependency and active AI engagement. Interviews were conducted virtually via Zoom, Google Meet, or phone, and lasted between 30 and 45 minutes each.

Each interview followed a structured 10-question guide covering four thematic sections. Section 1 (Questions 1–3) explored data access and sharing challenges: a specific instance when data access became a problem (Q1), datasets the startup knows exist but cannot access (Q2), and a full narrative account of any data-sharing experiences including trust dynamics, contract negotiations, and technical obstacles (Q3). Section 2 (Questions 4–6) addressed regulatory barriers: a real-world example of regulatory difficulties (Q4), any business opportunities turned down or product launches delayed due to regulatory concerns (Q5), and the impact of cross-border data rules with specific examples (Q6). Section 3 (Questions 7–8) covered the AI adoption journey: how the organisation started using AI/ML and what data sourcing and ownership challenges arose (Q7), and the most significant current barrier to adopting or scaling AI (Q8). Section 4 (Questions 9–10) captured recommendations and support needs: the single most impactful change the startup would make to the data or regulatory landscape (Q9), and the most useful legal templates or frameworks Digital Africa could develop (Q10).

Interview responses were lightly edited for readability and clarity while preserving the original substance and meaning of all responses. Specific client names or commercially sensitive details mentioned by interviewees have been omitted in accordance with confidentiality commitments.

Outreach Methodology and Reach

The survey was disseminated over a three-month outreach campaign using a multi-channel approach designed to maximise reach across the eight priority countries and four priority sectors. Outreach was conducted using standardised templates adapted across all messaging channels, inviting startup founders to help shape the future of AI legislation in Africa. All respondents were offered a complimentary copy of “The Data Protection Guide for Startups in Africa,” a 50+ page practitioner guide covering the Malabo Convention, the EU GDPR, and local data protection legislation across major startup hub nations in Sub-Saharan Africa.

The primary outreach channels were: direct messaging and contact enrichment via Lemlist, through which over 3,000 startup professionals across 300+ startups were contacted after sourcing emails and phone numbers via LinkedIn contact enrichment and direct LinkedIn outreach; and direct engagement through over 30 private WhatsApp groups serving African startup communities, founders, and ecosystem professionals, including founders, executive team members, heads of product, and heads of AI development.

The email campaign achieved an open rate of approximately 20% and a click-through rate to the survey form of approximately 2%, resulting in a survey drop-off rate of around 95% – a figure consistent with the broader challenge of engaging busy startup founders with lengthy online surveys absent strong incentives. Personal calls and direct messages via WhatsApp and LinkedIn proved substantially more effective at generating responses, though repeated follow-ups were required even with personalised communication.

This experience underscores a systemic insight: without strong, tangible incentives and personal outreach, voluntary participation from startup founders in structured research remains very difficult to sustain at scale.

Scope and Limitations

This study is explicitly scoped to four sectors (AgriTech, Climate Tech, Energy Tech, and Food Tech) which reflect the AI4Startups programme's thematic priorities. Startups operating in other sectors were excluded from the primary analysis, even where they participated in the survey, as their experiences may not reflect the specific data governance challenges in agriculture, food systems, climate, and energy.

The findings presented in this report are exploratory and qualitative in character. They reflect the experiences of a specific cohort of early-to-mid-stage African startups and should not be treated as statistically representative of the African tech ecosystem as a whole.

The study is designed to establish a baseline and generate actionable insights, with the expectation that annual repetition will build comparability over time. Potential sources of bias include self-reporting bias (mitigated by anonymity and the toolkit incentive), selection bias toward more digitally engaged startups (mitigated by broad multi-channel outreach), and recall bias in interview responses (mitigated by focusing on recent, specific experiences).

EXECUTIVE SUMMARY

This study documents the data governance and regulatory challenges facing African startups in the AgriTech, Climate Tech, Energy Tech, and Food Tech sectors across eight priority countries. Thirty-nine startups completed the quantitative survey, and twelve in-depth qualitative interviews were conducted across Ghana, Senegal, Côte d'Ivoire, Kenya, Rwanda, South Africa, Tanzania, and Botswana. The following five headline findings define the landscape:

- Data is mission-critical, but access remains severely constrained. 97% of respondents rated reliable data access as 'Somewhat Critical,' 'Critical,' or 'Very Critical' to their core operations. Yet the most commonly reported data access barrier was poor data quality, followed by unclear permissions or licensing. The gap between data dependency and practical data access is the defining tension of this cohort.
- AI adoption is active and accelerating but infrastructure, talent, and costs constrain scale. 47% of respondents are already running AI in production. The leading AI adoption barriers are the cost of AI tools, cloud services, and hardware, followed by data privacy concerns and concerns about the reliability of AI outputs. Unlike more general-purpose tech cohorts, regulatory uncertainty ranks lower than infrastructure and talent as an AI barrier in this sector-specific cohort.
- Data protection laws are broadly perceived as enabling rather than limiting – but compliance remains difficult. 72% of respondents said that data protection and sharing laws primarily help their work. However, the most difficult regulatory areas are data governance and internal policies, followed by data protection and privacy compliance. Cross-border transfer rules and data ownership rules are also frequently cited as difficult, reflecting the multi-country operating reality of this cohort.
- Data sharing is widespread but inadequately governed. 69% of respondents have shared data with another organisation. Among those who have, the most common challenges are data security concerns, lack of formal agreements, unclear data ownership, and different data protection standards. Informal agreements (meeting notes, email, verbal arrangements) are common, creating governance gaps that undermine the trust needed for sustained collaboration.
- The single most demanded support is better data-sharing platforms, followed by clearer laws and AI capacity building. Better data-sharing platforms (42%), clearer laws and regulations (39%), and technical training and AI capacity building (39%) are the top three support needs. Simplified legal templates and access to quality training data also feature prominently.

The most actionable gap identified by this study is the mismatch between the data-sharing activity that is already occurring and the governance infrastructure supporting it. Most startups are sharing data but without adequate legal frameworks, technical standards, or trust mechanisms. Addressing this gap through practical legal templates, regional data standards, and targeted capacity building is the clearest immediate priority for the AI4Startups platform.

SECTION 1: WHO RESPONDED

1.1 Geographic Distribution

The 39 in-scope survey respondents are distributed across all eight priority countries, with Tanzania and Côte d'Ivoire representing the largest shares of the cohort.

Country	Respondents	% of Total
Tanzania	8	21%
Côte d'Ivoire	8	21%
South Africa	7	18%
Rwanda	5	13%
Botswana	5	13%
Ghana	4	10%
Senegal	2	5%
Kenya	1	3%

The increased weight of Tanzania and Côte d'Ivoire in this updated cohort reflects the addition of five new respondents: Uji Moto Org (Tanzania), Khula (South Africa), Brastone (Côte d'Ivoire), Newform Foods (South Africa), and KTech Information Services (Kenya). The Francophone West Africa representation remains limited and should be a priority for expansion in the next survey round.

1.2 Sectoral Composition

The sectoral balance is broadly appropriate for a cohort centred on the agri-food and clean energy opportunity in Africa.

Sector	Respondents	% of Total
Food Tech	13	33%
AgriTech	11	28%
Energy Tech	10	26%
Climate Tech	5	13%

1.3 Startup Stage Breakdown

The cohort spans the full early-to-mid-stage spectrum. Seed-stage startups represent the largest single group, followed by Pre-seed and Bootstrapped. This stage diversity enables comparison of how data governance challenges evolve as companies scale.

Stage	Respondents	% of Total
Pre-seed	12	31%
Seed	13	33%
Bootstrapped	8	21%
Series A	4	10%
Series B+	2	5%

1.4 Cohort Summary

This is a multi-country, multi-sector cohort of predominantly early-to-mid-stage startups with high data dependency and active AI engagement. The concentration in agri-food and clean energy sectors and the spread across eight African countries makes this cohort well-suited to generate insights relevant to the AI4Startups programme's mandate. All eight priority countries are represented.

SECTION 2: DATA DEPENDENCY & ACCESS CHALLENGES

2.1 How Data-Critical Are These Startups?

Data criticality across the cohort is high. 38% of respondents described their core business as 'Very Critical' and 31% as 'Critical.' Only one respondent described data as not critical to their operations.

Response	Count	% of Total
Very critical – core business model depends on consistent data access	15	38%
Critical – lack of quality data significantly impacts delivery	12	31%
Somewhat critical – data improves but isn't essential	11	28%
Not critical	1	3%

2.2 The Data Access Gap

Despite high data dependency, barriers to access are pervasive. Respondents cited multiple challenges simultaneously. Poor data quality is by far the most commonly reported barrier, cited by 58% of respondents – a finding that reflects the particular challenge of agricultural, climate, and food systems data, where collection infrastructure is often underdeveloped and standardisation is inconsistent.

Data Access Barrier	Count	% of Respondents
Poor data quality	23	59%
Unclear permissions or licensing	16	41%
Technical barriers (format, integration)	11	28%
Data is too expensive to obtain	10	26%
Insufficient technical expertise / talent	9	23%
Insufficient quantity of available data	9	23%

Slow internal approval and data access processes	7	18%
Data is unavailable or doesn't exist	5	13%

These barriers cluster into three recurring themes: quality and availability failures (data exists but is too poor or scarce to be usable); legal and permission barriers (unclear licensing, data ownership ambiguity, and restricted government access); and infrastructure and cost barriers (expensive procurement, technical incompatibility, and the cost of sourcing quality agricultural and environmental data).

2.3 The Data Sharing Landscape

69% of respondents have shared data with another organisation – a substantially higher rate than might be expected given the stage distribution. Among those who have, the most commonly reported challenges were concerns about data security (most frequently cited), lack of formal agreements or contracts, unclear data ownership or rights, different data protection standards between parties, and lack of trust.

The prevalence of data sharing – combined with the absence of robust governance structures supporting it – is one of the most important findings of this study. Startups are already sharing data; the infrastructure to do so safely and sustainably is not keeping pace.

2.4 Stories From the Field: Data Access in Practice

Government Data Silos – Nutricom (Tanzania, Food Tech)

Nutricom needed official malnutrition data from Tanzania's Ministry of Health to design a nutritionally appropriate fortified porridge for school feeding programmes. Despite the data existing within government systems, the request process was slow, unclear, and ultimately unsuccessful.

"It feels like the data is there, but startups like ours are left outside the door."

– Nutricom, Tanzania

The team was forced to rely on fragmented NGO surveys, delaying the product launch by nearly six months and weakening their case with donors. This pattern – data that nominally exists but is practically inaccessible due to siloed systems and bureaucratic gatekeeping – recurs across multiple interviewees in this cohort.

Commercial Data Lock-In – Grainotech (Côte d'Ivoire, AgriTech)

Grainotech was unable to access soil fertility maps from the Ministry of Agriculture to validate precision farming recommendations for smallholder cooperatives. Satellite imagery and rainfall data from international agencies are locked behind expensive paywalls. Locally, crop yield records collected by government extension officers remain siloed.

"The barrier is less about availability and more about gatekeeping and lack of open data policies."

– Grainotech, Côte d'Ivoire

Trust as a Data Access Barrier – Swahili Honey (Tanzania, Food Tech)

Swahili Honey needed official honey production volume statistics across Tanzanian regions to forecast supply for export contracts. The data was not publicly available from the Ministry of Agriculture. An attempt to collaborate with a smallholder cooperative by sharing hive productivity data stalled because cooperative members feared the data could expose operational weaknesses or be used against them commercially. The barrier was not technical but relational.

Utility Data Restriction – Eazypay (Botswana, Energy Tech)

Eazypay sought detailed electricity consumption data from a local utility to design a prepaid energy solution. The data existed within the utility's systems but was inaccessible due to internal policies. Forced to rely on rough estimates, the product rollout was delayed and initial calibration was imprecise.

"The data existed, but we couldn't get access to it because of internal restrictions. That forced us to rely on rough estimates."

– Eazypay, Botswana

SECTION 3: AI ADOPTION

3.1 Where Are Startups on the AI Adoption Curve?

AI adoption in this cohort is active. 47% of respondents are already using AI in production, and an additional 10% are actively piloting or testing. A further 23% are planning implementation within six months. Only 20% report no current plans or uncertainty.

AI/ML Status	Count	% of Total
Yes, in production	18	46%
Piloting / Testing	4	10%
Planning to implement within 6 months	9	23%
No plans currently / Not sure	8	21%

3.2 Barriers to AI Adoption and Scale

Among respondents currently using or piloting AI, the top barriers to scaling were infrastructure and cost-related rather than purely regulatory.

AI Barrier	Count	% of AI-Active Respondents
AI tools, cloud services, and hardware costs	12	55%
Data privacy concerns	10	45%
Low customer demand for AI-based solutions	10	45%
Data procurement and licensing expenses	9	41%
Concerns about the reliability of AI outputs	9	41%
Difficulty integrating with existing systems	8	36%
Need for further research into AI applications	8	36%
Insufficient technical expertise / talent	7	32%

The dominance of infrastructure cost and output reliability as top barriers reflects the on-the-ground realities of deploying AI in agricultural and energy contexts, where ground-truth data for model validation is sparse, compute infrastructure is expensive, and customer willingness to pay for AI-driven services is still being established.

3.3 AI Adoption in Practice

Demand Prediction – Nutricom (Tanzania, Food Tech)

Nutricom built its first AI system to predict demand for fortified foods across regions, drawing on a mix of open-source nutrition data, climate information, and internal sales records.

“By using AI, we hoped to match production more closely to actual needs, reducing losses and improving access.”

– Nutricom, Tanzania

Solar Generation Optimisation – Gigawatt (Rwanda, Climate Tech)

Gigawatt started using AI to predict solar generation patterns in rural areas, optimising battery usage and reducing downtime for off-grid communities. Training data was drawn from open-source weather data and internal pilot installations.

No-Show Prediction – Dineplan (South Africa, Food Tech)

Dineplan trained its first AI model on anonymised reservation histories from Johannesburg and Cape Town to predict restaurant no-shows. By analysing booking patterns, party size, and weather, it achieved meaningful reductions in no-show rates – directly saving restaurant revenue. The talent scarcity problem subsequently prevented scaling the recommendation engine beyond the prototype stage.

Crop Yield and Market Trend Prediction – Brastone (Côte d’Ivoire, AgriTech)

Brastone began experimenting with AI to predict crop yields and detect market trends, drawing on farmer surveys and publicly available weather datasets. The primary ongoing barrier is data quality and granularity: satellite imagery is available but often too expensive or insufficiently detailed for smallholder plots.

“Without reliable, granular datasets, our AI models risk producing inaccurate recommendations.”

– Brastone, Côte d’Ivoire

SECTION 4: THE REGULATORY LANDSCAPE

4.1 Do Regulations Help or Hinder?

A striking finding from this cohort is the degree to which data protection laws are perceived as enabling rather than limiting. 72% of respondents said that data protection and sharing laws primarily help their work. Only 10% said laws primarily limit operations.

View	Count	% of Respondents
Primarily help	28	72%
Not sure	3	8%
No significant impact	4	10%
Primarily limit	4	10%

This finding warrants careful interpretation. For startups in agri-food and clean energy sectors that are actively seeking partnerships with governments, utilities, and international buyers, data protection laws can function as legitimising frameworks providing structure that builds partner confidence and enables formal collaboration. The positive view of regulation may also reflect the fact that many respondents have not yet encountered the full compliance burden at scale.

4.2 The Most Difficult Regulatory Areas

Despite the broadly positive view of regulation, specific compliance areas are widely experienced as difficult. Data governance and internal policies and data protection and privacy compliance are the top two difficult areas, each cited by roughly half of respondents. Cross-border transfer rules and data ownership are also significant pain points.

Regulatory Area	Count	% of Respondents
Data governance / Internal policies	19	49%
Data protection / Privacy compliance	18	46%
Cross-border data transfer rules	12	31%
Data ownership rules	11	28%

Data-sharing agreements	10	26%
Data localization requirements	6	15%
Consent management	5	13%
Security requirements	5	13%
AI/algorithmic transparency requirements	3	8%

The high ranking of data governance and internal policies above the often-cited data localisation and cross-border transfer rules suggests that the most immediate compliance burden for these startups is not international regulatory harmonisation but the ability to establish functional internal data governance systems. Many are sharing data and using AI without the internal policies and documentation frameworks needed to do so responsibly.

4.3 The Legal Capacity Landscape

Unlike broader cohorts where a quarter of respondents had no legal support at all, this sector-specific cohort shows stronger, though still uneven, legal capacity. Most respondents have access to some form of legal guidance, whether in-house or external. However, the quality and relevance of that guidance for data governance and AI specifically is likely highly variable. Critically, a meaningful minority – including SafiRun, East Africa Foods, and Shumba – continue to operate without any legal support whatsoever.

4.4 Regulatory Challenges in Practice

Cross-Border Data Conflicts – Multiple Interviewees

Cross-border data transfer challenges emerged consistently across the interview cohort, regardless of country or sector. The pattern is similar in each case: a startup seeking to expand operations encounters conflicting data protection requirements between its home country and the destination market.

“Tanzania’s data protection rules conflicted with Kenya’s requirements, especially around consent and anonymization. We had to redesign our analytics pipeline to strip out identifiable information, which added costs and slowed down our ability to scale.”

– **Nutricom, Tanzania**

“Scaling from Botswana into neighbouring countries has been slowed by different interpretations of data protection laws. Some require local storage, others allow cloud. This creates technical duplication and higher costs.”

– **Brastone, Côte d’Ivoire**

This pattern, repeated across Gigawatt (Rwanda to Uganda), TechShelta (Ghana to Côte d’Ivoire), Kofa (Ghana to Nigeria), Swahili Honey (Tanzania to Kenya), and Eazypay (Botswana to South Africa) indicates a systemic structural problem with cross-border data governance that cannot be resolved startup by startup.

Overlapping Permits for Innovation Hardware (Multiple Interviewees)

A second recurring regulatory theme is the challenge of importing innovation hardware including smart meters, IoT sensors, EV components, beekeeping equipment, and food processing machinery into markets without clear regulatory categorisation for emerging technology.

“When we imported smart meters, customs required multiple overlapping permits. The lack of a clear regulatory framework for IoT devices meant weeks of paperwork and storage fees at the border.”

– Gigawatt, Rwanda

This pattern was reported by Gigawatt, Open Access Energy, Plentify, Everlectric, Eazypay, Kofa, TechShelta, Grainotech, Swahili Honey, and Dineplan covering all four sectors and multiple countries. It constitutes one of the three recurring regulatory challenge themes identified by this study.

Regulatory Gaps Around Carbon, Traceability, and Emerging Models

A third recurring theme is the regulatory gap around emerging commercial models including carbon credit markets, digital traceability systems, and peer-to-peer energy trading that are being developed by startups faster than regulatory frameworks can accommodate them.

“Ghana’s regulatory framework for carbon trading was still under development. Without clarity, we couldn’t risk exposing customers to legal uncertainty. This delay cost us a partnership with an international buyer.”

– Kofa, Ghana

“Tanzania’s food safety certification rules were unclear about how digital traceability systems should be recognized. Without clarity, we couldn’t risk rolling out the feature.”

– Nutricom, Tanzania

In each case, the startup was commercially ready to launch but legally unable to do so not because a rule prohibited the activity, but because no clear rule existed to govern it. Regulatory gaps are as commercially damaging as regulatory overreach.

SECTION 5: WHAT STARTUPS NEED

5.1 The Support Needs Landscape

Respondents were asked what would most help their startup use and share data more effectively. The results reveal a pragmatic, platform-oriented set of priorities.

Support Need	Count	% of Respondents
Better data-sharing platforms	16	41%
Clearer laws and regulations	15	38%
Technical training / AI capacity building	15	38%
Training and capacity building (general)	14	36%
Access to quality training data	13	33%
Partnerships with data providers	12	31%
Simplified legal templates	11	28%
Ethical AI guidelines	11	28%
More affordable access to data	10	26%
Regional data standards (African standards)	10	26%

The top priority — better data-sharing platforms — points directly to the infrastructure gap identified in Section 2. Startups are sharing data, but without adequate technical infrastructure to do so reliably, securely, and at scale. This is an actionable and concrete platform development priority.

The equal ranking of clearer laws and AI capacity building reflects the dual nature of the challenge: startups need both the regulatory clarity to operate confidently and the technical capability to make effective use of data and AI tools.

5.2 What Startups Recommend: Voices From the Interviews

Regional Open Data Hubs – The Dominant Recommendation

The single most consistent recommendation across all 12 interviews regardless of sector or country was the creation of sector-specific, regionally accessible open data repositories. Every interviewee pointed to government or commercial data that exists but is practically inaccessible, and called for structured mechanisms to make that data available to startups.

“If we could change one thing, it would be the creation of a regional open food and nutrition data hub. A single, standardized repository where startups, researchers, and policymakers could access consumption, production, and health datasets without gatekeeping.”

– **Nutricom, Tanzania**

“We would create a pan-African open climate data hub, a single repository of energy, weather, and environmental datasets. It would eliminate gatekeeping, reduce duplication, and accelerate innovation across the continent.”

– **Gigawatt, Rwanda**

The convergence on this recommendation across Food Tech (Nutricom, Swahili Honey), Climate Tech (Gigawatt, Kofa, TechShelta), Energy Tech (Open Access Energy, Plentify, Everlectric, Eazypay), and AgriTech (Grainotech, Brastone) is striking. It represents the clearest and most actionable policy recommendation to emerge from this study.

Standardised Data-Sharing Agreement Templates

The second most consistent recommendation was for standardised, practically usable data-sharing agreement templates covering data ownership, liability, consent, permitted use, and cross-border applicability.

“Every time we try to collaborate, whether with farmer cooperatives, research institutions, or government agencies, we spend months negotiating contracts around liability, ownership, and compliance. A clear, practical agreement would save us enormous time and legal fees.”

– **Nutricom, Tanzania**

“A plug-and-play data-sharing agreement template would be invaluable. Negotiating contracts with each restaurant group currently takes months and legal fees.”

– **Dineplan, South Africa**

Clearer Regulatory Categorisation for Emerging Technology Hardware

Several interviewees recommended that governments clarify the regulatory categorisation of emerging technology hardware including smart meters, IoT sensors, and EV components to eliminate the overlapping permit problem documented in Section 4.

“A framework that clearly defines data ownership, balances liability, aligns with Ghana’s Data Protection Act, and anticipates cross-border flows would save us enormous time and legal costs.”

– **TechShelta, Ghana**

SECTION 6: CONCLUSIONS AND RECOMMENDATIONS

6.1 Three Recurring Regulatory Challenge Themes

In line with the Lot 1 mandate, this study identifies the following three recurring regulatory challenge themes across both the survey data and the interview narratives:

- Theme 1: Cross-border data governance fragmentation. Startups attempting to scale across African borders consistently encounter conflicting data protection requirements that require costly technical redesign, slow expansion timelines, and increase operational costs. This challenge affects all four sectors and every country pair examined in the interview cohort. It cannot be resolved at the individual startup level, it requires harmonised regional frameworks.
- Theme 2: Regulatory gaps around emerging commercial models. Startups in carbon markets, digital traceability, peer-to-peer energy trading, and AI-assisted food safety are commercially ready to launch products that existing regulatory frameworks do not adequately govern. The result is not prohibition but legal uncertainty which is commercially equivalent. Startups choose not to proceed rather than risk undefined liability.
- Theme 3: Government data inaccessibility. Across all four sectors, the data most needed by startups: malnutrition statistics, grid load data, soil fertility maps, emissions records, utility usage data, is held by governments and utilities that lack the open data infrastructure, API frameworks, or policy mandates to share it. This is the most fundamental structural barrier to AI adoption in this cohort.

6.2 Key Conclusions

- Data is mission-critical, but the access infrastructure does not match the need. The most commonly cited data access barrier is poor data quality, affecting 59% of respondents. Government and utility data that startups in agri-food and clean energy sectors depend on remains largely inaccessible, forcing reliance on expensive third-party sources or fragmented self-collected data.
- AI adoption is real and growing, but the constraints are infrastructural and financial rather than primarily legal. Infrastructure cost, output reliability, and customer market readiness are the dominant AI barriers in this cohort. Legal templates will help, but cannot substitute for investment in affordable compute access and AI capacity building.
- Data protection laws are broadly legitimising in this cohort but internal governance capacity is the gap. The most difficult regulatory area is data governance and internal policies, not cross-border rules or localisation requirements. Startups need help building internal data governance systems, not just navigating external regulations.
- Data sharing is widespread and inadequately governed. 69% of respondents have shared data. Among those who have, security concerns and lack of formal agreements are the top challenges. Standardised templates and data-sharing platforms could directly address this gap, and demand for both is high.
- The consensus recommendation across all sectors and countries is open, standardised, regional data repositories. Every interview cohort regardless of sector, country, or stage identified inaccessible government or commercial data as their primary constraint and called for open data hubs as the single most impactful change.

6.3 Recommendations for Policymakers and Regulators

The following recommendations are directed at national and regional regulators and are grounded in the cross-border complexity findings and regulatory difficulty assessments throughout this report. They are framed as practical, near-term opportunities rather than long-term systemic reform proposals.

Develop a harmonised cross-border data governance framework for priority sectors

The African Union's Data Policy Framework provides a strong foundation, but sector-specific protocols for agricultural data, energy system data, and food safety data are needed to enable startups to scale across borders without redesigning data infrastructure for each market. Concretely, this means establishing agreed definitions of what constitutes "personal data" in agricultural and energy contexts, harmonising consent requirements for secondary data use, and creating a single cross-border data transfer permit recognised across ECOWAS and EAC member states. The EU's adequacy decision mechanism under the GDPR offers a useful precedent: a similar recognition mechanism between African countries would dramatically reduce the legal and technical cost of cross-border expansion for startups.

Establish open data mandates for government-held agricultural, environmental, and energy datasets

National data protection laws should include mandatory provisions for API-accessible public interest data, starting with the categories most requested by startups: soil data, malnutrition statistics, grid load data, emissions records, and crop production records. Open data mandates should specify machine-readable formats, regular update schedules, and non-discriminatory access terms. Where governments cite privacy concerns as the reason for restricting access, anonymisation and aggregation standards should be developed collaboratively with the startup community to identify what level of detail can safely be made public. Kenya's Open Data Initiative and Rwanda's national data portal offer partial precedents worth building upon in other priority countries.

Create clear regulatory pathways for emerging commercial models

Carbon credit markets, digital food traceability systems, pay-as-you-go energy, and peer-to-peer energy trading all require regulatory frameworks that are currently absent or incomplete in most priority countries. Regulatory sandboxes – already deployed in Kenya, Rwanda, and South Africa in fintech contexts should be extended explicitly to cover clean energy, agritech, and food safety innovations. Each sandbox should include time-bounded safe harbour provisions that protect participating startups from retroactive liability while frameworks are being developed. Ghana's ongoing development of a carbon trading framework and Tanzania's evolving digital food safety certification regime are examples where sandbox engagement could accelerate both regulatory design and startup viability.

Harmonise regulatory categorisation of innovation hardware

Smart meters, IoT sensors, EV components, and agricultural technology devices should have clear, single-pathway import and operational licensing in each country eliminating the overlapping permit problem that is currently causing weeks of delays across all four sectors. Regional customs harmonisation bodies such as the East African Community Customs Union and ECOWAS Trade Liberalisation Scheme should develop a dedicated product classification category for "digital agriculture and clean energy devices" that allows streamlined clearance across member states. Short-term, national revenue authorities should issue interpretive guidance clarifying which existing HS codes apply to smart meters, IoT agricultural sensors, and EV charging infrastructure.

6.4 Recommendations for the AI4Startups Platform

The following recommendations are directed at Digital Africa and the AI4Startups platform team, grounded in the support needs and practitioner recommendations documented throughout this report.

Prioritise five legal template types based on interview evidence

Based on the recurring data-sharing scenarios described across all 12 interviews, the following five template types should be developed as the highest-priority Lot 2 deliverables: (1) an Agricultural Data Sharing Agreement covering farm-level data shared with financial institutions, cooperatives, and buyers, with explicit clauses on farmer consent, data minimisation, and commercial use restrictions; (2) an Energy System Data Collaboration Framework for utility and smart meter data shared with agritech, fintech, and government partners, including liability caps for inaccurate AI predictions based on utility-provided data; (3) a Climate and Environmental Data Access Agreement covering satellite, emissions, and weather data sourced from government and international agencies, with provisions for academic and commercial use differentiation; (4) a Food System Traceability Data Agreement for supply chain and nutritional data shared with buyers and regulators, with specific clauses on data verification standards and regulatory recognition; and (5) a Cross-Border Data Transfer Protocol, a jurisdiction-adaptable overlay template that can be appended to any of the above, specifying which country's law governs, how cross-border transfers are structured, and what anonymisation standards apply.

Each template should be available in both English and French, include plain-language summaries of key clauses, and be accompanied by a negotiation guide identifying the clauses most commonly contested in startup-to-institution negotiations.

Develop a data-sharing platform or brokerage mechanism as the highest-priority infrastructure investment

Better data-sharing platforms is the single most demanded support need in this cohort (41%). A platform that enables structured, governed data exchange between startups, governments, and research institutions with built-in consent management, audit trails, and role-based access controls would address the governance gap more effectively than templates alone. The platform should draw on existing open data infrastructure where it exists (Kenya's Open Data Portal, Rwanda's National Data Hub) and work to extend interoperability across priority countries. Critically, the platform should be sector-configured: agricultural datasets require different access controls and anonymisation standards than energy grid data, and a one-size-fits-all approach will not serve this cohort's needs.

Build sector-specific data access guides for all four priority sectors

Plain-language guides explaining what government data exists in each priority country, how to formally request it, what licensing applies, what fees (if any) are involved, and what to do when access is denied would immediately reduce the time and cost startups spend navigating government data bureaucracies. These guides should be updated annually, include sample request letters and follow-up templates, and flag known access pathways that have worked for other startups. The guides should cover AgriTech data needs (soil, crop yield, weather, land records), Food Tech needs (nutrition statistics, food safety certification, traceability standards), Energy Tech needs (grid load, outage histories, meter data), and Climate Tech needs (emissions data, carbon registry access, meteorological records).

Advocate for AI inference credits and affordable compute access for African startups

The cost of AI tools, cloud services, and hardware is the single largest barrier to AI adoption in this cohort. Group rate agreements with major cloud providers (AWS, Google Cloud, Microsoft Azure), modelled on existing academic research credit programmes, should be negotiated at the programme level and made available to all AI4Startups cohort members.

Regional AI compute credit programmes similar to the EU's EuroHPC Joint Undertaking, should be advocated for at the AU level, with Digital Africa positioned as a convener and early adopter. In the near term, the AI4Startups platform should curate and actively promote existing startup credit programmes (AWS Activate, Google for Startups, Microsoft for Startups) that many respondents are unaware of.

Commission a follow-up validation workshop to test draft templates

Per the Lot 1 methodology, a virtual or hybrid validation workshop presenting these findings and initial draft templates to representative startups from across the eight priority countries would ensure the tools developed under Lot 2 are grounded in actual startup needs. The workshop should include structured feedback sessions on each of the five recommended template types, with particular attention to the clauses that startups find most difficult to negotiate in practice. Findings from the workshop should be incorporated into the final template versions before publication.

APPENDIX A: IN-SCOPE SURVEY RESPONDENTS

The following 39 startups are included in this study's primary analysis. All operate in one of the eight priority countries and one of the four priority sectors. [Click here to access survey data.](#)

#	Startup	Sector	Stage	Country
1	Umm Pod	AgriTech	Pre-seed	Tanzania
2	Dineplan	Food Tech	Bootstrapped	South Africa
3	My Bid Food	Food Tech	Bootstrapped	South Africa
4	Kauai	Food Tech	Bootstrapped	South Africa
5	Loopa	AgriTech	Bootstrapped	Rwanda
6	GET IT	AgriTech	Series A	Rwanda
7	PulaSafe	Climate Tech	Seed	Botswana
8	WeFly Agri	AgriTech	Series A	Côte d'Ivoire
9	Cool Lion	Climate Tech	Series B+	Côte d'Ivoire
10	Jaza	Energy Tech	Pre-seed	Tanzania
11	Solar Taxi	Energy Tech	Seed	Ghana
12	East Africa Foods	Food Tech	Series B+	Tanzania
13	Green Agro Valley	AgriTech	Series A	Côte d'Ivoire
14	Open Access Energy	Energy Tech	Seed	South Africa
15	Plentify	Energy Tech	Series A	South Africa
16	Kofa	Climate Tech	Seed	Ghana
17	Everlectric	Energy Tech	Seed	South Africa
18	TechShelta	Climate Tech	Seed	Ghana
19	Heny	Food Tech	Seed	Ghana
20	Swahili Honey	Food Tech	Pre-seed	Tanzania
21	Nutricom	Food Tech	Pre-seed	Tanzania
22	SafiRun	Food Tech	Bootstrapped	Rwanda
23	Grainotech	AgriTech	Seed	Côte d'Ivoire

24	Agritech 4 Africa	AgriTech	Seed	Côte d'Ivoire
25	Shumba Ltd.	Energy Tech	Pre-seed	Botswana
26	Eazypay	Energy Tech	Pre-seed	Botswana
27	ARED	Energy Tech	Bootstrapped	Rwanda
28	Gigawatt	Climate Tech	Seed	Rwanda
29	MazaoHub	AgriTech	Pre-seed	Tanzania
30	ClimaCare	Climate Tech	Seed	Tanzania
31	Cashew Coast	Food Tech	Pre-seed	Côte d'Ivoire
32	Afrikamart	Food Tech	Seed	Senegal
33	Solarbox	Energy Tech	Pre-seed	Senegal
34	Choice Tec Enterprise	AgriTech	Seed	Botswana
35	Uji Moto Org	Food Tech	Pre-seed	Tanzania
36	Khula	AgriTech	Pre-seed	South Africa
37	Brastone	AgriTech	Seed	Côte d'Ivoire
38	Newform Foods	Food Tech	Pre-seed	South Africa
39	KTech Information Services	Climate Tech	Bootstrapped	Kenya

APPENDIX B: INTERVIEW PARTICIPANTS

The following 12 startups participated in qualitative interviews for this study. [Click here to access interview data.](#)

#	Startup	Sector	Country	Key Interview Themes
1	Nutricom	Food Tech	Tanzania	Government data inaccessibility; cross-border conflicts; demand prediction AI; open data hub advocacy
2	Gigawatt	Climate Tech	Rwanda	Grid data access barriers; import permit overlaps; solar AI; cross-border Rwanda-Uganda friction
3	Grainotech	AgriTech	Côte d'Ivoire	Soil and satellite data access; Cdl-Ghana cross-border conflicts; pest detection AI
4	Open Access Energy	Energy Tech	South Africa	Eskom data restriction; POPIA compliance; peer-to-peer energy regulation gap
5	Kofa	Climate Tech	Ghana	EPA data access; carbon credit regulatory gap; Ghana-Nigeria cross-border friction
6	Dineplan	Food Tech	South Africa	POPIA compliance burden; reservation data sharing; no-show prediction AI; talent scarcity
7	Brastone	AgriTech	Côte d'Ivoire	Agricultural yield data access; farmer data sharing trust; cross-border localisation conflicts
8	Plentify	Energy Tech	South Africa	Municipal grid data access; POPIA compliance; energy trading regulation gap; demand AI
9	Everlectric	Energy Tech	South Africa	Eskom data restriction; EV regulation gaps; SA-Namibia cross-border friction; EV charging AI
10	TechShelta	Climate Tech	Ghana	Meteorological data access; carbon credit regulatory gap; Ghana-Cdl cross-border friction
11	Swahili Honey	Food Tech	Tanzania	Honey production data inaccessibility; cooperative trust barriers; traceability regulation gap
12	Eazypay	Energy Tech	Botswana	Utility data access; blockchain energy regulation gap; Botswana-SA cross-border friction